Genotoxicity and Sub-acute Toxicity Assessment of methanolic *Cordyceps militaris* (L.) Fr. Extract in Swiss Albino Mice

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ABSTRACT

Ethnopharmacological relevance: Cordyceps militaris, a traditional medicinal fungus, parasitizes the intestines of lepidopteron pupae or larvae, predominantly during the winter, and undergoes fruiting in the summer or autumn. Compounds extracted from *C. militaris* have demonstrated a broad spectrum of pharmacological effects, including antioxidant, anti-tumor, anti-metastatic, snit-inflammatory, antiviral, anti-diabetic, and various others.

Aim of the study: Herein, our study aimed at elucidating the acute, sub-acute toxicity, and genotoxicity profiles of *C. militaris* methanolic extract following oral administration in Swiss albino mice, representing the inaugural comprehensive exploration of the toxicological and safety profiles of *C. militaris*.

Materials and Methods: Prior studies have predominantly focussed on its biological activities rather than its toxicity. Acute oral toxicity study was conducted at 2000 and 5000 mg/Kg doses of *C. militaris* over a 14-day period. Three groups of mice (six of either gender): two groups were administered 300 mg/mL of the extract consecutively for 14 and 28 days; one group served as a control. Mice were monitored for their body weight and behavioural changes once daily. Hematological, serum biochemical, histopathological, histomorphometric, seminal parameters, and mutagenic investigations were performed post-treatment period.

Results: Acute oral toxicity study at 2000 and 5000 mg/Kg doses revealed no signs of toxicity, with an LD_{50} value surpassing 5000 mg/Kg. No instances of mortality reported. Body and organ weights were altered significantly, with no behavioural changes. Hematological analysis illustrated a marked upsurge in RBCs, Hb, HCT, WBC, PLT, MPV, RDW, PCT, PDW, along with a substantial increase in neutrophil percentage with concomitant decrease in lymphocyte percentage after 28-day. Liver enzyme tests showed elevated ALP, while renal enzymes displayed changes in serum CRE and BUN levels. Genotoxicity profile and histopathological assessments of the liver, spleen, testis, and ovary manifested no remarkable irregularities, except for mild renal toxicity. Seminal parameters including sperm concentration, motility and testosterone levels displayed noteworthy increase.

Conclusions: The study sheds light on the potential risks and safety considerations associated with *C. militaris*-based medicinal products. These findings establish a foundation for further investigations and the refinement of dosage optimization in the application of *C. militaris,* with the aim of mitigating any potential adverse effects.

Keywords: Cordyceps militaris; Histopathology; Kidney; Mutagenic; Sperm; Sub-acute doses

1. Introduction

Natural resources have historically been employed as remedies or preventative measures for a diverse array of ailments owing to their potent therapeutic benefits and cost-effectiveness. Individuals resort to natural substances believing that they pose minimal risks and do not entail adverse consequences (Ha et al., 2018). Medicinal plant species incorporate an extensive repertoire of secondary metabolites, some of which exhibit remarkable intricacy (Huang et al., 2021). Nonetheless, in most instances, the utilization of such plant-based remedies for the treatment of specific illnesses lacks scientific understanding or confirmation of their toxicological effects (Koriem et al., 2019).

The acute and sub-acute toxicity tests are crucial for assessing drug safety, and concurrently understanding their mode of action. Within the realm of chemical production, handling, and usage, acute and sub-acute systemic toxicity studies are utilized to disclose hazards and mitigating associated risks (Li et al., 2020). For centuries, a diverse array of mushrooms has traditionally been used in numerous countries for health benefits and treating ailments. “Cordyceps”, a mushroom of traditional Chinese origin, is categorized within the phylum Ascomycota, sometimes referred to as "Sac fungi" (Yang et al., 2020). With an estimated 750 identified species, Cordyceps stands as the largest and most varied genus within the Cordycipitaceae family in terms of species diversity, morphology, and host variability (Das et al., 2010). All recognized species of this well-known parasitic fungus of the ascomycetes group act as endoparasitoids, predominantly inhabiting insects and other arthropods, while a few display remarkable host specificity (Islam et al., 2021). This survival strategy has facilitated the host's defences to produce an assortment of distinctive secondary metabolites, showing potential for the advancement of innovative pharmaceuticals (Quy et al., 2019).

Since its inception as a therapeutic agent in traditional Chinese medicine many centuries ago, cordyceps remained relatively unnoticed until the year 1993, when a breakthrough occurred among Chinese athletes who consumed *C. sinensis* (CS) tonic into their regimen, consequently achieving unprecedented athletic feats (Hirsch et al., 2017). Following this milestone, cordyceps and its derivatives as sustenance, remedy, and tonic gradually captivated global interest (Zhang et al., 2020). Owing to their rarity and exorbitant costs linked to their natural cultivation, the majority of cordyceps species are presently artificially cultivated, showcasing robust bioactivities akin to the wild ones. *Cordyceps militaris* (L.) Fr. (CM) emerges as a distinguished artificially cultivated mushroom species renowned for its significance in traditional healing practices, extensively employed as an alternative to CS in health supplements (Lin et al., 2021; Nxumalo et al., 2020; Yu et al., 2006). CM encompasses a wide array of bioactive constituents, including cordycepin, polysaccharides, adenosine, amino acids, cordycepic acid, sterols, all of which demonstrate noteworthy pharmacological effects (Wu et al., 2024). In the realm of traditional Chinese medicine, CM extracts (both aqueous and alcoholic) are prevalent for the purpose of boosting immunity, assuaging lung discomfort, restoring kidney function to address conditions like hyperglycemia, hyperlipidemia, renal dysfunction, hyperuricemia, hepatic ailments, respiratory diseases, cough and asthma, among other therapeutic uses (Fung et al., 2011; Han et al., 2010; Kim et al., 2014; Nxumalo et al., 2020; Yong et al., 2016; Yue et al., 2008; Zhu et al., 2013). Traditionally, CM has also garnered considerable usage as a raw medicinal substance and a traditional health-promoting food in regions of East Asia (Lee et al., 2017). Numerous investigations have documented the multifaceted biological properties displayed by CM, including its anticancer, antioxidant, anti-diabetic, immunomodulatory, antiviral, and anti-inflammatory properties, among others (Han et al., 2011; Quy et al., 2019; Rao et al., 2010; Wada et al., 2017; Yu et al., 2015).

To the greatest extend of our comprehension; investigations pertaining to the chemical or toxicological properties of the methanolic extract derived from CM fruiting bodies have yet to be undertaken. It is imperative to assess the safety profile of this particular mushroom. Within the confines of this research, we are investigating any plausible immediate and prolonged toxic consequences that may arise from the methanolic extract of CM fruiting bodies. To achieve this, several parameters have been taken into account including hematological indices, biochemical markers, histopathological and histomorphometric assessment of the organs, semen analysis, as well as mutagenic assessments like sperm abnormality tests and chromosomal aberration study.

2. Materials and methods

2.1. Sample preparation

The sample, *Cordyceps militaris* 3936, was procured from Cosmic Cordyceps farm, Haryana, and was originally sourced from IMTECH, Chandigarh. Subsequently, it was cultured under controlled laboratory facilities, as outlined by Dutta et al. (2024). Following cultivation, the fruiting bodies were harvested, and processed using 70% methanol. Methanol, being a versatile polar solvent, exhibits high solubility towards a broad spectrum of compounds, encompassing both polar and non-polar substances as evidenced in TLC profile utilizing chloroform, methanol, and distilled water as the mobile phase (Supplementary Figure 1). This property makes it effective in extracting various bioactive compounds, as illustrated in the HPLC profile, conducted using a DAD detector, Signal A (254.0 nm/Bw:4.0 nm Ref 360.0 nm/Bw:100.0 nm) for adenosine, and Signal B (210.0 nm/Bw:4.0 nm Ref 360.0 nm/Bw: 100.0 nm) for cordycepin (Supplementary Figure 2). Additionally, its low boiling point eases the evaporation process and concentration of the extract, thereby avoiding excessive exposure to heat that could potentially deteriorate subtle compounds. Subsequently, the methanolic *C. militaris* extract (CME) underwent filtration, evaporation at 40ºC using rotary evaporator, and preserved at 4ºC for further experimental use.

2.2. Experimental animals

Ten to twelve-week-old Swiss albino mice of both sexes, weighing 25-30 grams, were maintained in-house under standardized laboratory conditions with a 12-hour light/dark cycle, relative humidity 50-65%, and stable room temperature of 20-22ºC. The mice were acclimatized for 15 days prior to commencement of the experiment and had standard food and unrestricted access to water.

2.3. Acute Oral Toxicity

Acute toxicity studies adhered to Organization for Economic Co-operation and Development (OECD) guidelines 425 (OECD, 2022), using Swiss albino mice of both sexes. Five mice received a single oral dosage of CME at 2000 mg/Kg body weight (B.W.) subsequent to an overnight fasting with water *ad libitum*, and monitored closely for 24 hours. Surviving mice received a higher dose of 5000 mg/Kg B.W. for 14 days.

2.4. Sub-Acute Oral Toxicity

The sub-acute oral toxicity study followed OECD 407 guidelines, accessing comprehensive safety parameters (OECD, 2008). Thirty-six animals (both sexes, n=6, each sex per group) were randomly assigned to three groups for hematological, biochemical, histopathological, and mutagenic analyses. The extract was deemed safe up to 2000 mg/Kg B.W. in mice. To determine the effective dose, typically $1/5^{th}$ to $1/10^{th}$ of the lethal dose is taken. Therefore, experimental animals administered oral dose of 300 mg/Kg B.W. methanolic CME daily, and grouped into three viz., Group I: Administered 1X PBS alone (Vehicle control)

Group II: Administered methanolic CME at 300 mg/Kg B.W. per day for 14 days

Group III: Administered methanolic CME at 300 mg/Kg B.W. per day for 28 days

Feeding was commenced 2 to 3 hours post-dosing, with animals monitored for any signs of mortality and toxicity for a month. The animal's conduct was tracked daily for 3 hours until the treatment period concluded. Daily observations included fur condition, body colour, eye appearance, behaviour, breathing, abdominal changes, urine/faeces, and aggression.

2.5. Body weight and Organ weight of mice

Mice's body weight was recorded daily throughout the treatment. Organ weight, both absolute and relative, was examined after 14- and 28-day treatment period, post-sacrifice. The relative organ weight was calculated considering the respective body weight at that time by using the following formula:

$$Relative\ organ\ weight\ (\%) = \frac{Organ\ weight}{Body\ weight} \times 100$$

2.6. Hematological indices

Fresh blood samples were obtained via retro-orbital sinus puncture and placed in EDTA (BD K2E) containing vacutainer tubes for complete blood count (CBC) analysis following 14- and 28-day CME treatment. The diverse CBC parameters including hemoglobin (Hb), red blood corpuscles (RBC) count, white blood corpuscles (WBC) count, hematocrit (HCT), platelet count (PLT), plateletcrit (PCT), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), mean platelet volume (MPV), red cell distribution width (RDW), platelet distribution width (PDW), and differential leucocyte count (DLC), were measured using an Automated Hematology Analyzer (Biobase BK-3200) (Yang et al., 2019).

2.7. Biochemical assay

Blood was collected in an SST® gel and clot activator vacutainer tube (Becton and Dickinson, Franklin Lakes, NJ, USA) for serum biochemical analysis. Serum was isolated by centrifuging the tube at 1500 rpm for 10 minutes at ambient temperature. Liver and kidney functions indices such as, aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), Blood Urea Nitrogen (BUN) and Creatinine, were measured using an Automated Clinical Biochemical Analyzer (Idexx VetTest 8008) (Rodriguez-Lara et al., 2019).

2.8. Seminal analysis and serum testosterone assay

Following blood sampling, semen was retrieved from the caudal epididymis and vas deferens of mice using a syringe, diluted in phosphate buffer, and incubated at 37ºC for seminal evaluation. Total sperm concentration and motility were recorded (Saleem et al., 2020). The enumeration of spermatozoa was conducted using hemocytometer and expressed in millions per millilitre (x 10^6 per mL) in accordance with the guidelines set forth by W.H.O. The images were captured using Labomed Lx 500 microscope and analyzed with Motic Image Plus Version 2.0 software. Percentage of sperm motility was computed using the formula: (total number of sperm – number of inactive sperm)/ total number of sperm x 100%. The serum testosterone concentration was estimated using a commercial kit (Bioassay Technology Laboratory kit, Cat No. E0260Mo).

2.9. Histopathological and Histomorphometric studies

The mice were euthanized at the end of the experiment and organs like liver, kidney, spleen, testis, and ovary were excised, fixed in 10% neutral buffered formalin, and processed post 14 and 28-day CME treatment. Tissues underwent histological processing including ethanol dehydration, clearing, paraffin embedding, sectioning (5 µm), and Hematoxylin and Eosin (H & E) staining (Malatesta, 2016). Upon histomorphometric assessment, all measurements were performed using calibrated microscope and Motic Image Plus 2.0 software in adherence to standard guidelines (Farokhi et al., 2007; Fazelipour et al., 2012; Jaji et al., 2019; Ortega-Martinez et al., 2021; Rezigalla, 2022; Weibel et al., 1969). Each group's minimum of 20 slides were assessed and scored for the degree of changes.

2.10. Mutagenic analysis

2.10.1. Sperm morphology assay

Male mice from various groups were sacrificed after 14 and 28-day CME treatment. Both the caudal epididymis was excised and meticulously macerated to release spermatozoa, which were then spread onto clean slides, air-dried, fixed in absolute ethanol, and stained with 1% aqueous Eosin-Y. Five hundred spermatozoa from each mouse were evaluated for any deviations in head and tail morphology, following established experimental guidelines (Wyrobeck and Bruce, 1975).

2.10.2. Chromosomal aberration study

Chromosomal aberration analysis was performed following Ansari et al. (2019). Briefly, mice were administered intraperitoneal colchicine injections (4 mg/mL B.W.) two hours prior to sacrifice to induce mitotic arrest at the metaphase chromosomes. Bone marrow was aspirated from the femur bone by flushing it with 0.075M KCl solution (pH 7.0), centrifuged at 1000 rpm, and pellet fixed in Carnoy's solution. On a clean, grease-free, chilled slide, few drops of this suspension was pipetted from a distance, burned on a flame, air dried, and stained with Giemsa. Aberrations including chromatid breaks and gaps, exchanges, sister chromatid union and chromosomal fragments, were scored by monitoring few hundred high-quality metaphase spreads.

2.11. Statistical analysis

The results were expressed as the mean ± S.E.M. (standard error mean). Data were analyzed for normality test using Shapiro-Wilk's W test and was further analyzed for One-way analysis of variance (ANOVA) followed by Bonferroni post-hoc multiple comparisons test when ANOVA indicated significant. All differences were considered significant at the 95% confident level.

3. Results

3.1. Acute Oral Toxicity study

To ascertain the drug's level of toxicity, an acute oral toxicity study was carried out. No mortality was observed at 2000 and 5000 mg/Kg B.W. of mice. Till 14-day of observation, all animals were found to be normal and showed no clinical signs of significant morphological or behavioral changes, suggesting that the LD_{50} of the extract exceeds 5000 mg/Kg B.W., indicating that it is safe and non-toxic in nature.

3.2. Behavioral study of mice

Typical behavior of the mice following CME treatment remained unchanged with normal fur condition, body color, breathing, faeces, urine color, and eye appearance. They showed no aggression, drowsiness, abdominal abnormalities, or hostility. Importantly, all mice survived till the culmination of the study. Their behavior within the initial three hours post-treatment is as shown in Table 1.

Table 1 Behavioral observation of mice during first three hours post-CME treatment.

Sl. No.	Parameters	Observations
1	Fur condition	Normal
2	Eye dullness	No
3	Body color	No change
4	Swelling	No
5	Breathing	Normal
6	Color of faeces and urine	Normal
7	Lethargic	No
8	Abdominal distention	No
9	Abdominal contraction	No
10	Aggressive behavior	No

3.3. Body weight and Organ weight of mice

Male mice treated with CME showed body weights (g) fairly similar to that of control group; however 28-day treated female mice exhibited a significant decrease (One-way ANOVA: $F_{2,6} = 9.30$, $p \leq 0.05$) (Fig. 1). Kidney absolute weight (g) decreased significantly in females (One-way ANOVA: $F_{2,6} = 7.81$, $p \leq 0.05$) in both 14 and 28-day treated mice while spleen showed a significant increase after 28-day treatment (One-way ANOVA: $F_{2,6} = 6.69$, $p \leq 0.05$). Furthermore, relative organ weights (%) showed significant changes in kidney (One-way ANOVA; Male: $F_{2,6} = 27.62$, $p \leq 0.05$ and Female: $F_{2,6} = 19.79$, $p \leq 0.05$), spleen (One-way ANOVA;

Male: $F_{2,6} = 9.34$, $p \leq 0.05$ and Female: $F_{2,6} = 43.20$, $p \leq 0.01$), and testis (One-way ANOVA: $F_{2,6} = 15.69$, $p \leq 0.05$).

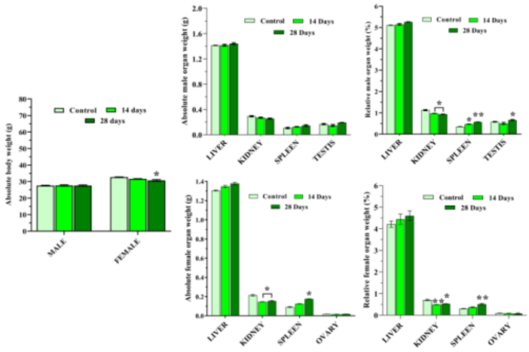

Fig. 1. Body weight and organ weight (absolute and relative) of mice post CME treatment. Data are expressed as mean ± S.E.M (n = 3); One-way ANOVA followed by Bonferroni post hoc test for multiple comparisons. Results were considered statistically significant at $*p \leq 0.05$; $**p \leq 0.01$ as compared to control group.

3.4. Hematological assay

Hematological profile of untreated and CME treated mice is illustrated in Fig. 2. CME administration led to a significant increase in RBC count (One-way ANOVA; $F_{2,6} = 18.85$, $p \leq 0.05$), Hb (One-way ANOVA; $F_{2,6} = 6.78$, $p \leq 0.05$), HCT (One-way ANOVA; $F_{2,6} = 8.31$, $p \leq 0.05$), MPV (One-way ANOVA; $F_{2,6} = 30.41$, $p \leq 0.05$), PCT (One-way ANOVA; $F_{2,6} = 11.17$, $p \leq 0.05$), PDW (One-way ANOVA; $F_{2,6} = 41.06$, $p \leq 0.05$), WBC count (One-way ANOVA; $F_{2,6} = 16.23$, $p \leq 0.05$), and platelets (One-way ANOVA; $F_{2,6} = 69.01$, $p \leq 0.01$), compared to the control group. However, the differential leucocyte count did not exhibit any noteworthy changes, except for neutrophils percentage, which showed a significant increase (One-way ANOVA; $F_{2,6} = 62.22$, $p \leq 0.01$), with concomitant decrease in lymphocytes percentage (One-way ANOVA; $F_{2,6} = 21.62$, $p \leq 0.01$), upon treatment. Nevertheless, no significant differences ($p > 0.05$) were observed between the control and treated group in terms of MCV, MCH, and MCHC.

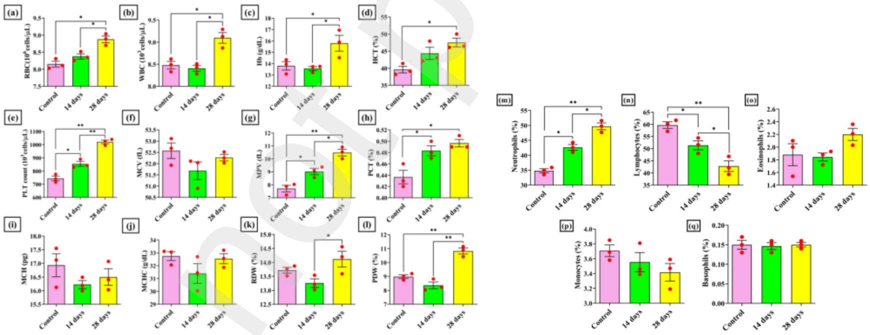

Fig. 2. Hematological indices elucidate complete blood count (a-l) and differential leucocyte count (m-q) of Swiss albino mice after 14 and 28-day methanolic CME treatment. Data are expressed as mean ± S.E.M (n = 3); One-way ANOVA followed by Bonferroni post hoc test for multiple comparisons. Results were considered statistically significant at $*p \leq 0.05$; $**p \leq 0.01$ as compared to control group.

3.5. Biochemical assay

Based on the outcome of the liver function test, CME treatments noticeably changed the serum levels of AST (One-way ANOVA; $F_{2,6} = 0.20$) and ALT (One-way ANOVA; $F_{2,6} = 1.99$), but are not statistically significant ($p > 0.05$). Nevertheless, compared to 28 days, ALP levels increased notably during the first 14 days of treatment (One-way ANOVA; $F_{2,6} = 10.49$, $p \leq 0.05$) as shown in Fig. 3. Additionally, renal biomarkers such as BUN (One-way ANOVA; $F_{2,6} = 26.33$, $p \leq 0.05$) and creatinine (One-way ANOVA; $F_{2,6} = 26.10$, $p \leq 0.001$) showed a significant decrease in 14- and 28-day treatment, respectively, compared to control group.

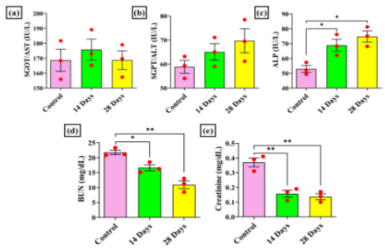

Fig. 3. Biochemical indices of swiss albino mice in the sub-acute toxicity study following CME treatment. Data are expressed as mean ± S.E.M (n = 3); One-way ANOVA followed by Bonferroni post hoc test for multiple comparisons. Results were considered statistically significant at *$p \leq 0.05$; **$p \leq 0.01$ as compared to control group.

3.6. Seminal analysis and Testosterone assay

The sperm concentration (x 10^6/mL) (One-way ANOVA; $F_{2,6} = 16.25$, $p \leq 0.05$), motility (%) (One-way ANOVA; $F_{2,6} = 19.88$, $p \leq 0.05$) and serum testosterone concentration (ng/dL) (One-way ANOVA; $F_{2,6} = 9.83$, $p \leq 0.05$) significantly increased after 28-day CME treatment as compared to control (Fig. 4). The sperm concentration increased from 52.13 ± 3.60 to 73.20 ± 5.03 (x 10^6/mL); sperm motility increased from 58 ± 7 to 84 ± 5 %; serum testosterone level increased from 6.65 ± 2.42 to 9.71 ± 5.10 ng/dL after CME treatment period of 28 days.

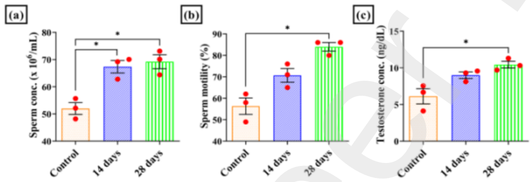

Fig. 4. Effects of CME on (a) sperm concentration (x 10^6/mL), (b) sperm motility (%), and (c) serum testosterone level (ng/dL) of male mice. Data are expressed as mean ± S.E.M (n = 3); One-way ANOVA followed by Bonferroni post hoc test for multiple comparisons. Results were considered statistically significant at *$p \leq 0.05$ as compared to control group.

3.7. Histopathological and Histomorphometric observations

3.7.1. Microscopic alterations of liver

Histopathological study of liver sections of control group showed typical hepatic architecture with normal hepatocyte size and shape (Fig. 5a). The portal triad consisting of portal vein, hepatic artery and bile ductules are distinctly marked (Fig. 5b). Slight changes has been observed in the CME treated groups such as mild congestion of central vein (Fig. 5c), endothelial disruption in central and portal veins (Fig. 5c, m), and perivascular fibrosis (Fig. 5h). Sinusoids displayed edema and mild inflammatory symptoms manifested by slight accumulation of Kupffer cell throughout the tissue (Fig. 5d, e, j). Fig 5 (d and e) showed the presence of bi- and tri-nucleated hepatocytes structured in distinct cords. These hepatocytes frequently demonstrated pyknosis, karyorrhexis and karyolysis (Fig. 5d, e, f, j, l). In the sinusoidal space, an area of coagulated necrosis was seen, followed by infiltration (Fig. 5k). Following treatment, hepatic arteries and bile ductules exhibited slight distortion and dilation (Fig. 5f, l, m, n). Figure 5 and Table 2 provide specifics regarding the liver section's histological characteristics and karyometric parameters.

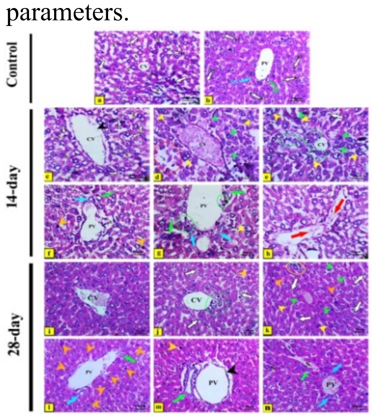

Fig. 5. Photomicrographs of H & E stained liver sections of (a, b) control and (c-n) CME treated mice. White arrow: KC; blue arrow: HA; green arrowhead: karyolysis, karyorrhexis; green arrow: BD; green circle: mononuclear infiltration surrounding CV; yellow arrowhead: bi- and tri-nucleated hepatocytes; yellow circle: area of coagulative necrosis infiltrated by inflammatory cells; black arrowhead: congested CV; red arrow: perivascular fibrosis; red circle: sinusoidal edema; orange arrowhead: pyknotic nuclei; CV: central vein; PV: portal vein; S: sinusoids; KC: kupffer cells; HA: hepatic artery; BD: bile duct (magnification 400x), line bar in µm.

3.7.2. Microscopic alterations of kidney

Histopathological and Histomorphometric features of kidney from control and treated groups are illustrated in Fig. 6 and Table 3. The control group showed normal histological structure with the renal cortex displaying renal corpuscles having normal diameter comprising glomerular capillaries encased in Bowman's capsule (Fig. 6a). The proximal convoluted tubules (PCT) were lined with pyramidal cells with narrow lumina, while the distal convoluted tubules (DCT) were lined with cuboidal cells with large lumina. Renal medulla comprised of collecting tubules, Loop of Henle and interstitial blood capillaries lined by cuboidal epithelium (Fig. 6b). In contrast to the control group, renal tissue sections of the 14-day treated group showed notable histopathological changes such as marked glomerular congestion (Fig. 6c), sub-capsular epithelial desquamation with pyknotic nuclei (Fig. 6d), glomerular mesanglial hyper-cellularity (Fig. 6e), vacuolation in sub-capsular space (Fig. 6f), and accumulation of reticulated casts in renal tubular lumina (Fig. 6g). Some renal corpuscle showed significant reduction in glomerular diameter accompanied by hypotrophy and degeneration with loss of contents (Fig. 6h, Table 3). While few others displayed glomerular atrophy with nuclear pyknosis of intraglomerular cells (Fig. 6i). Certain tubules revealed shedding of the cellular cytoplasm in the tubular lumina leading to tubular edema (Fig. 6i), desquamation (Fig. 6j), and dilation (Fig. 6n). Furthermore, renal medulla exhibited interstitial leukocyte and inflammatory cells infiltration (Fig. 6k, m). In the medulla, perivascular leukocyte infiltration and perivascular edema were also noted with mononuclear cell infiltration (Fig. 6k, l). Conversely, 28-day treated mice unveiled further changes such as conjoint renal corpuscles (Fig. 6o), polymorphonuclear infiltration in glomerular space (Fig. 6t), degeneration and edema of renal tubules (Fig. 6x), wide spacing of tubules with epithelial necrosis (Fig. 6y). It also revealed a marked decrease in the size of renal corpuscles (Fig. 6q). Later, tubular infiltration was found to be associated with glomeruli loss (Fig. 6v).

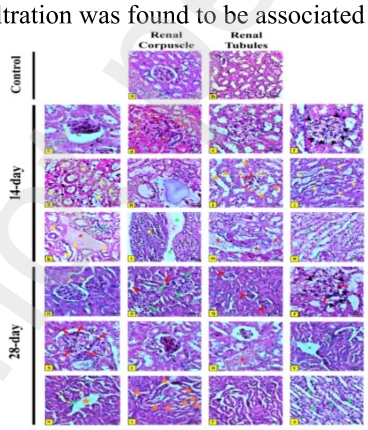

Fig. 6. Photomicrographs of H & E stained kidney sections of (a, b) control and (c-z) CME treated mice. Yellow arrowhead: desquamation; yellow arrow: conjoint renal corpuscle; yellow star: accumulation of reticular casts; red arrowhead: decrease in renal corpuscles; red arrow: pyknotic nuclei; red star: inflammatory infiltration; blue arrowhead: tubular degeneration; blue arrow: degenerated renal corpuscle; orange star: tubular edema; green arrowhead: tubular dilation; green star: mononuclear cell interstitium; green cone: perivascular edema; black arrowhead: vacuolation in sub-capsular space; black circle: wide spacing of tubules with epithelial necrosis; ∗ : loss of glomeruli; G: glomerulus; BC: Bowman's capsule (magnification 400x), line bar in µm.

3.7.3. Microscopic alterations of spleen

Histopathological studies of the spleen in control group revealed the typical appearance of splenic parenchyma which is primarily composed of two functional zones: the white pulp and the red pulp (Fig. 7a). The white pulp consists of periarteriolar lymphoid sheath (PALS) clustering around T-lymphocyte-dominated central arterioles. The germinal center is located in the middle of the white pulp, which is predominantly made up of B-lymphocytes (Fig. 7a, b). In contrast, the venous sinuses that the arteriolar blood opens into are formed by the red pulp. The marginal zone, on the other hand, marks the interface between the white and red pulp (Fig. 7b). Upon 14-day treatment, the spleen showed very few alterations like minimal hypocellularity of the PALS region which gave a distinct appearance of germinal centers (Fig. 7d, f), slight vacuolation and hemosiderosis in the red pulp (Fig. 7e, g). Conversely, 28-day treatment exhibited disrupted splenic architecture with indistinct red and white pulp showing little atrophy (Fig. 7h, i). Compared to red pulp, white pulp appeared more diffused and less well-defined (Fig. 7j, k). There has been a negligible decrease in the white pulp's area (mm^2) and perimeter (mm). However, after receiving treatment, the red pulp area (mm^2) showed a significant increase. Table 4 depicted a detailed illustration of these morphometric variations of the spleen.

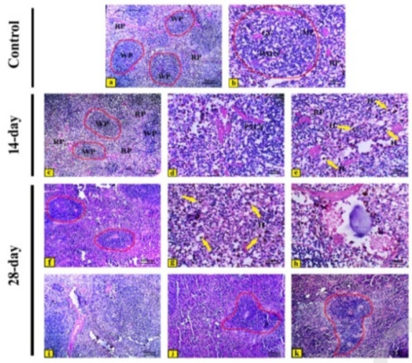

Fig. 7. Photomicrographs of H & E stained spleen sections of (a, b) control and (c-k) CME treated mice. Yellow arrow denotes hemosiderosis (H) in the red pulp (e, g). Area encircled red line depicted white pulp region (a,b,c,f) having diffused white pulp (j,k). RP: red pulp; WP: white pulp; PALS: periarteriolar lymphoid sheath; MZ: marginal zone; GC: germinal center (magnification 100x, 400x), line bar in µm.

3.7.4. Microscopic alterations of testis

Histological examination of the testis of control mice showed well-defined seminiferous tubules containing spermatogonia, spermatocytes, spermatids, sertoli cells, spermatozoa in their lumen (Fig. 8a, b). Male reproductive hormone is produced by a small number of cell clusters called Leydig cells, which are located around the seminiferous tubules (Fig. 8a, c). The size and mean area of the seminiferous tubules of 28-day treated group showed a notable increase than control group (Fig. 8h, Table 5). Larger the mean area more mature is the seminiferous tubule. However, the germinal epithelium was found slightly disorganized with few cavities (Fig. 8h); the number of seminiferous tubules persisted alike and revealed an irregular shape. Interstitial space exhibited minor increase post 28-day treatment (Fig. 8i). Conversely, the histology of the Leydig cells in the treated groups was somewhat analogous to that of the control mice (Fig. 8c, f, i).

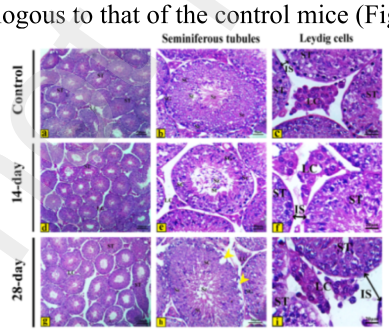

Fig. 8. Photomicrographs of H & E stained testis sections of (a, b, c) control and (d-i) CME treated mice. The germinal epithelium surrounding seminiferous tubules was observed to be mildly disordered with a limited number of cavities, as indicated by yellow arrowhead (h). Black double arrow indicates slight increase in interstitial space (f, i) (magnification 100x, 400x), line bar in µm.

3.7.5. Microscopic alterations of ovary

Histological interpretations of the ovary of control mice displayed normal primordial, primary, secondary, pre-antral, antral, Graafian, and pre-ovulatory follicles. The follicles when mature are covered by layers of cells called theca externa and theca interna. Corpus luteum showed general appearance of luteal cells. The secondary oocyte is lined by zona pellucida layer directly surrounded by corona radiata, cumulus oophorus and granulosa cells (Fig. 9a-f). Following 28-day CME treatment, slight vacuolation in interstitium with regressing primordial follicles (Fig. 9s), degenerated granulosa cells in tertiary follicle (Fig. 9v), and loss of corona radiata and cumulus oophorus surrounding oocyte was observed (Fig. 9w). Moreover, a minor shift in the ooplasm to the periphery (Fig. 9x) with slight loss of luteal cells from the corpus luteum (Fig. 9z) was perceived. Table 6 illustrates the histomorphometric examinations of the ovarian follicles which displayed a significant increase in the size of oocytes of primordial, primary, secondary and Graafian follicles post treatment ($p > 0.05$).

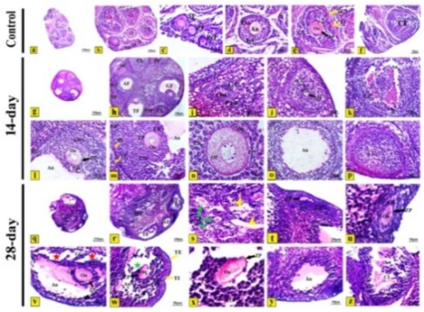

Fig. 9. Photomicrographs of H & E stained ovary sections of (a-f) control and (g-z) CME treated mice. Green arrow: regressing primordial follicle; green star: degenerated granulosa cells in Graafian follicle; yellow arrowhead: vacuolation in interstitium; yellow arrow: theca externa and theca interna; red star: degenerated granulosa cells in tertiary follicle; black arrow: zona pellucida. PrF: primordial follicle; PF: primary follicle; SF: secondary follicle; TF: tertiary follicle; GF: Graafian follicle; CL: corpus luteum; AF: antral follicle; An: antrum; Oo: oocyte; OP: ooplasm; ZP: zona pellucida; TI: theca interna; TE: theca externa; GC: granulosa cells; CR: corona radiata; CO: cumulus oophorus (magnification 40x, 100x, 400x), line bar in µm.

3.8. Sperm morphology assay

The results from the assessment of sperm morphology are visually depicted in Fig. 10. The heat map elucidates a comprehensive overview of the diverse categories of sperm abnormalities observed in both the head and tail portions (Fig. 10B). Interestingly, it has been observed that out of the 21 types of anomalies, sperm head exhibited maximum number of variations compared to the tail. The group subjected to a 28-day treatment displayed a greater range of irregularities in sperm morphology, while the midpiece was slightly distorted after a 14-day treatment. Nevertheless, the incidence of total abnormal sperm in the treatment groups did not significantly deviate from that of the control group ($p > 0.05$).

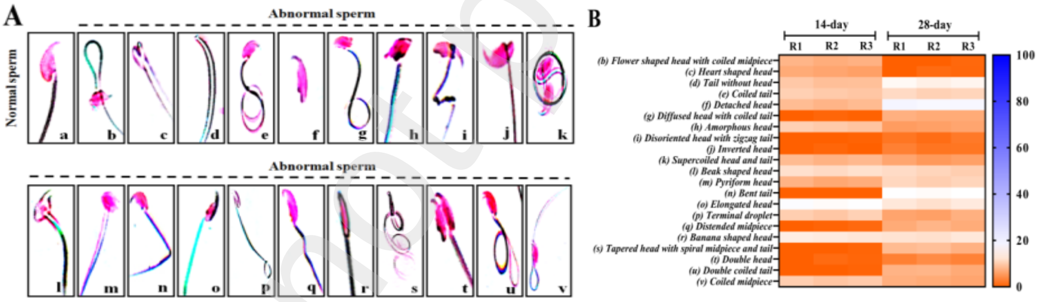

Fig. 10. (A) Types of morphological defects of sperm in (a) control and (b-v) treated groups. (B) Percentage of sperm abnormalities following 14 and 28-day CME treatment as compared to control.

3.9. Chromosomal aberration study

The types of chromosomal abnormalities were discovered to be moderately higher subsequent to CME treatment during the corresponding time period of 28 days; however their percentage is almost negligible as compared to control. In the current study, 500 evenly spread out metaphase chromosomes from each cohort were scrutinized to determine the impact of treatment on bone marrow cells (Fig. 11). In accordance with the current findings, the chromosomes of the mouse bone marrow exhibited both structural and numerical anomalies post treatment. Structural chromosomal aberrations encompass chromatid deletions (D), chromatid fragments (F), centric fusion (CF), end-to-end joining (E-E), ring or sister chromatid union (SCU), single chromatid break (SCB), iso-arm fragmentation (ISAF), single chromatid gap (SCG), iso-chromatid gap (ISCG), iso-chromatid break (ISCB), centric gap (CG), acentric fragment (AF), and dicentric chromosome (DC). Polyploidy is a type of numerical aberration that was persuaded at 300 mg/Kg B.W. of methanolic CME as illustrated in Fig. 11(o). Following 14-day treatment, the incident of aberrant metaphase spread of chromosomes increased to 6.01 ± 1.20 %, as compared to 4.97 ± 1.71 % in control group. Conversely, 28-day treated group displayed 7.17 ± 2.07 %. However, all these values are non-significant ($p > 0.05$).

Table 2 Mean values of histomorphometric observations of the liver in control and experimental groups.

Parameters	Treatment groups				
	Control	14 days	28 days	F(2,6)	*p* value
Longest axis (µm), D	266.23 ± 4.33	260.8 ± 8.02	233.06 ± 5.11**	24.88	0.0012
Shortest axis (µm), d	123.01 ± 6.60	111.58 ± 7.20*	93.10 ± 3.44**	58.88	0.0011
Mean axis (µm): $M = (D \times d)^{1/2}$	180.25 ± 1.10	169.13 ± 4.30*	155.28 ± 3.50**	26.58	0.0010
D/d ratio	2.10 ± 5.02	2.27 ± 4.03	2.39 ± 7.06*	6.31	0.0337
Perimeter (µm): $P = (\pi/2) \times [1.5 (D + d) – M]$	620.11 ± 15.13	608.77 ± 11.80	551.01 ± 19.20**	16.69	0.0035
Area (µm²): $A= \pi \times M^2/4$	25403.1 ± 16.12	23560.6 ± 11.46*	19234.1 ± 10.80**	26.16	0.0013
Volume (µm³): $V= \pi/6 \times M^3$	3032460 ± 28.03	2614902 ± 30.10*	2103681 ± 25.20**	29.13	0.0021
V/A ratio (µm)	119.12 ± 5.01	107.82 ± 7.12*	98.23 ± 2.20**	17.12	0.0016
Shape factor: $F= 4 \times \pi \times A/P^2$	0.82 ± 0.02	0.80 ± 0.01	0.79 ± 0.02	4.01	0.6110
Contour index: $I= P/(A)^{1/2}$	3.90 ± 2.10	3.92 ± 1.30	3.93 ± 2.01	1.57	0.2671
Eccentricity: $E= (D + d)^{1/2} \times (D - d)^{1/2} /D$	0.88 ± 0.02	0.90 ± 0.01	0.91 ± 0.04	1.19	0.3603

Data are expressed as mean ± S.E.M (n = 3); Data were analyzed by One-way ANOVA followed by Bonferroni post hoc test for multiple comparisons. Results were considered statistically significant at $*p \leq 0.05$; $**p \leq 0.01$ as compared to control group.

Table 3 Mean values of histomorphometric observations of the kidney in control and experimental groups.

Parameters	Treatment groups				
	Control	14 days	28 days	F (2,16)	*p* value
Total no. of glomeruli/mm²	15 ± 2	14 ± 3	15 ± 3	1.10	0.3160
Diameter of glomeruli (µm)	64 ± 3.12	51 ± 2.33*	37.8 ± 3.40**	33.11	0.0006
Renal corpuscle diameter (µm)	80 ± 1.91	64.8 ± 2.10*	54.4 ± 4.10**	50.78	0.0002
Proximal convoluted tubule					
Shortest diameter (µm), d	37.03 ± 1.22	34.18 ± 4.11	28.72 ± 6.10*	5.513	0.0443
Longest diameter (µm), D	51.11 ± 2.17	45.81 ± 3.01	39.18 ± 7.03*	13.31	0.0062
Small circular area (SCA) (µm²)	965.30 ± 4.01	917.53 ± 9.11	647.82 ± 15.13**	44.50	0.0003
Large circular area (LCA) (µm²)	1993.10 ± 9.30	1671.14 ± 7.38**	1208.08 ± 13.11***	169.4	0.001
Elliptical area (EA) (µm²)	1475.20 ± 7.50	1310.30 ± 11.06	916.09 ± 16.10**	145.1	0.0019
Total area (TA) (µm²)	1313.44 ± 21.03	1201.09 ± 30.03	997.47 ± 25.42**	171.2	0.0020
Total perimeter (TP) (µm)	188.10 ± 4.22	173.90 ± 6.01	159.21 ± 3.81**	36.14	0.0005
Distal convoluted tubule					

Shortest diameter (µm), d	29.13 ± 1.12	26.06 ± 1.20	21.81 ± 2.47*	14.56	0.005
Longest diameter (µm), D	37.03 ± 2.07	34.55 ± 1.11	31.18 ± 1.60*	11.34	0.0092
Small circular area (SCA) (µm^2)	660 ± 31.51	558.22 ± 15.10*	393.59 ± 22.31**	314.2	0.0030
Large circular area (LCA) (µm^2)	1050 ± 15.30	937.53 ± 18.23*	763.55 ± 11.22**	218.2	0.01
Elliptical area (EA) (µm^2)	852.20 ± 4.13	723.43 ± 2.66	584.09 ± 5.19**	166.1	0.0060
Total area (TA) (µm^2)	901.03 ± 2.34	771.55 ± 4.00*	615.90 ± 1.21**	131.9	0.0031
Total perimeter (TP) (µm)	141.11 ± 1.31	128.77 ± 0.09*	106.27 ± 1.10**	174.2	0.0056
Thick Loop of Henle					
Shortest diameter (µm), d	30.02 ± 0.02	27.33 ± 0.11	20.33 ± 0.08**	43.16	0.0003
Longest diameter (µm), D	38.11 ± 2.10	34.88 ± 1.72	31.55 ± 1.04*	12.83	0.0068
SCA (µm^2)	688.01 ± 12.50	536.63 ± 19.30*	394.61 ± 25.40**	191.8	0.005
LCA (µm^2)	1071.37 ± 14.20	955.52 ± 10.11*	781.08 ± 15.40**	227.3	0.003
EA (µm^2)	872.20 ± 1.80	718.69 ± 0.59	560.76 ± 1.22**	506.9	0.01
Total area (TA) (µm^2)	866.23 ± 2.36	735.33 ± 5.50	607.35 ± 4.03**	412.3	0.01
Total perimeter (TP) (µm)	147.08 ± 0.04	130.55 ± 0.70*	102.83 ± 1.60**	161.8	0.01
Thin Loop of Henle					
Shortest diameter (µm), d	15.12 ± 0.20	13.4 ± 0.40	13.25 ± 0.19*	303.1	0.0669
Longest diameter (µm), D	24.23 ± 0.11	19.6 ± 0.50	17 ± 0.10*	416.3	0.0420
SCA (µm^2)	163.20 ± 7.40	141.02 ± 10.30*	137.88 ± 9.41**	12.63	0.0071
LCA (µm^2)	425.12 ± 8.11	301.71 ± 5.20*	226.08 ± 10.17**	311.2	0.006
EA (µm^2)	253.22 ± 2.30	206.27 ± 3.09	176.91 ± 5.05*	135.3	0.0213
Total area (TA) (µm^2)	293.22 ± 2.06	251 ± 1.27	182 ± 1.10*	241.8	0.0036
Total perimeter (TP) (µm)	84.13 ± 0.20	77 ± 0.11	60.42 ± 1.10*	12.23	0.0074
Collecting duct					
Shortest diameter (µm), d	25.64 ± 0.02	24.27 ± 0.60	22.87 ± 0.10	1.31	0.11
Longest diameter (µm), D	73.55 ± 4.29	69.12 ± 1.20	66.08 ± 3.70	7.11	0.10
SCA (µm^2)	493.12 ± 7.05	464.02 ± 5.60	441.79 ± 11.50	11.03	0.53
Total area (TA) (µm^2)	1801.23 ± 11.30	1548 ± 18.59**	1301.75 ± 9.20***	342.1	0.001
Total perimeter (TP) (µm)	432 ± 3.44	304.75 ± 2.01*	265.75 ± 3.11**	181.2	0.0056

Data are expressed as mean ± S.E.M (n = 3); One-way ANOVA followed by Bonferroni post hoc test for multiple comparisons. Results were considered statistically significant at *$p \leq 0.05$; **$p \leq 0.01$; ***$p \leq 0.001$ as compared to control group.

Table 4 Mean values of histomorphometric observations of the spleen in control and experimental groups.

Parameters	Treatment groups				
	Control	14 days	28 days	F (2,16)	*p* value
White pulp area (mm^2)	0.219 ± 0.01	0.211 ± 0.05	0.203 ± 0.02	1.504	0.291
White pulp perimeter (mm)	2.021 ± 0.02	1.814 ± 0.04	1.726 ± 0.20	1.182	0.306
Red pulp area (mm^2)	0.206 ± 0.20	0.239 ± 0.10*	0.396 ± 0.10**	310.1	0.0026
Red pulp perimeter (mm)	2.101 ± 0.03	2.243 ± 0.01*	2.706 ± 0.20**	56.83	0.0011

Data are expressed as mean ± S.E.M (n = 3); One-way ANOVA followed by Bonferroni post hoc test for multiple comparisons. Results were considered statistically significant at $*p \leq 0.05$; $**p \leq 0.01$ as compared to control group.

Table 5 Mean values of histomorphometric observations of the testis in control and experimental groups.

Parameters	Treatment groups				
	Control	14 days	28 days	F (2,16)	*p* value
Leydig cell area (μm^2)	107.33 ± 1.40	98.61 ± 1.11	104.17 ± 1.30	1.19	0.13
Leydig cell perimeter (µm)	37.8 ± 2.10	33.50 ± 2.30	35.6 ± 0.50	1.51	0.11
Leydig cell diameter (µm)	14.13 ± 1.10	15.03 ± 2.11	11.36 ± 1.20	2.34	0.30
Seminiferous tubule area (μm^2)	1471.06 ± 9.21	1822.20 ± 13.58**	2308.3 ± 16.30***	78.51	0.001
Seminiferous tubule perimeter (µm)	498.10 ± 12.10	541.03 ± 7.11	689.45 ± 5.60**	17.93	0.0029
Seminiferous tubule diameter (µm)	129.20 ± 3.40	137.63 ± 2.10	158.70 ± 2.50*	5.98	0.0372

Data are expressed as mean ± S.E.M (n = 3); Data were analyzed by One-way ANOVA followed by Bonferroni post hoc test for multiple comparisons. Results were considered statistically significant at $*p \leq 0.05$; $**p \leq 0.01$; $***p \leq 0.001$ as compared to control group.

Table 6 Mean values of histomorphometric observations of the ovary in control and experimental groups.

Parameters	Treatment groups				
	Control	14 days	28 days	F (2,16)	*p* value
Total number of healthy follicles	226 ± 10.10	233 ± 11.09	241 ± 8.10**	24.16	0.0046
Oocytes diameter (µm)					
Primordial follicle	30.2 ± 0.40	42.8 ± 0.21*	55.13 ± 0.30**	141.2	0.0060
Primary follicle	34.8 ± 0.11	49.2 ± 0.10*	61.03 ± 0.50**	110.1	0.0021
Secondary follicle	51.2 ± 0.21	58.2 ± 0.13*	73.03 ± 0.47**	64.28	0.0074
Graafian follicle	138.6 ± 0.15	160.10 ± 0.20*	175.21 ± 0.13**	38.12	0.0013

Data are expressed as mean ± S.E.M (n = 3); Data were analyzed by One-way ANOVA followed by Bonferroni post hoc test for multiple comparisons. Results were considered statistically significant at $*p \leq 0.05$; $**p \leq 0.01$ as compared to control group.

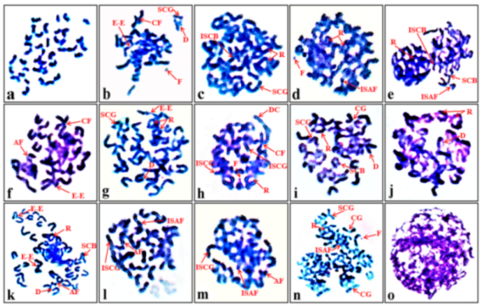

Fig. 11. Photomicrographs of metaphase spread from mice bone marrow cells showing chromosomes from control (a) and CME treated groups (b-o). Different forms of chromosomal anomalies observed during the study were D: deletion, F: chromatid fragment, R: AF: acentric fragment, CF: centric fusion, CG: centric gap, DC: dicentric chromosome, E-E: end-end joining, SCG: single chromatid gap, SCU: sister chromatid union or ring, SCB: single chromatid break, ISCB: iso-chromatid break, ISAF: iso-arm fragmentation, ISCG: iso-chromatid gap.

4. Discussion

At the present time, there is a well-established understanding of the pharmacological impacts of medicinal plants, although knowledge regarding the potential toxicity of their active components remains inadequate (Benrahou et al., 2022). Initial stages of substance development often involve acute toxicity tests on rodent models to evaluate toxicity (Yang et al., 2019). Recent research illustrated that the daily administration of 2000 or 5000 mg/Kg B.W. of methanolic CME did not induce mortality or behavioural changes in mice, aligning with the findings on aqueous CME indicating a no-observed-adverse-effect level (NOAEL) exceeding 4000 mg/Kg B.W. per day in Sprague-Dawley (SD) rats (Jhou et al., 2018). Furthermore, the sub-acute toxicity evaluation of methanolic CME over 28 days revealed no altered behaviour or fatalities in mice.

Changes in body weight have served as a key metric for assessing the detrimental impacts of medications and chemicals (Vahalia et al., 2011). The average body weights of treated and control groups do not manifest any significant differences, except for 28-day treated female mice. A parallel outcome was witnessed in a study on SD rats using aqueous CME for 90 days (Jhou et al., 2018). Additionally, a statistically significant decrease in the absolute kidney weight was evident in female mice post 28-day treatment. This aligns with Raina et al. (2015), where organ weights reflect animal health status. Given their diverse functions within the human body, herbal products may pose threats to vital organs like kidneys, liver, and pancreas. Having said that, our study discovered certain discernible changes in the organs, as indicated by the outcomes derived from the assessment of organ weights in both treated and control groups.

Our study reveals that oral CME administration at 300 mg/Kg B.W. significantly increased RBC, WBC, Hb, HCT, PLT, MPV, PCT, RDW, and PDW levels, with no significant changes in MCV, MCH, and MCHC. Notably, CME-intoxicated mice showed altered neutrophil and lymphocyte percentages, aligning with the study by Jhou et al. (2018). In contrast, our study revealed a dose-dependent rise in WBC count and MPV, which could be attributed to the immunomodulatory function of CM (Liu et al., 2016). This immunomodulatory role might be reinforced by the presence of adenosine and cordycepin, as identified in the HPLC profile of methanolic CME (Supplementary Figure 2) and described by Zhu et al. (2013). Though the precise mechanisms underlying the elevation in haematological values induced by CM remain uncertain, it is plausible that the observed improvement in haematological values is associated with enhancing host immunity, ultimately improving host survivability.

Assessment of hepatic injury in toxicological research typically entails evaluating serum biochemical markers like ALT, AST, and ALP, indicative of enzyme release due to membrane disruption (Zhao et al., 2017). Our study noted a slight, non-significant increase in ALT and AST post-CME intervention. Conversely, ALP levels significant rose compared to the control group. This result finds support in a study on methanolic extract of *Erodium guttatum* demonstrating elevated levels of ALT and AST when contrasted with ethanolic and aqueous extracts (Benrahou et al., 2022). Correspondingly, BUN and creatinine decreased significantly post-CME. This corroborated with Choi et al. (2020) which showed BUN concentration reduced significantly at 300 mg/Kg of ethanolic CME in a 12 weeks study. However, Suwannasaroj et al. (2021) reported conflicting creatinine outcomes in Wistar rats.

Histopathological analysis of mice organs provides microscopic insights into potential toxicity. CME administration did not elicit any necropsy or histopathological alterations in liver, spleen, testis, and ovary (Fig. 5, 7, 8, 9). Liver histology displayed normal architecture (Fig. 5), albeit with mild changes like congestion of CV, endothelial disruption, pyknotic nuclei, activated kupffer cells, and slight sinusoidal edema. These nonspecific liver changes, coupled with a slight rise in ALP levels, hint at minimal treatment-related liver injury. Nevertheless, it is hypothesized that this outcome might reflect normal physiological responses rather than direct CME effects, which validate the findings from cordycepin treatment in rat (Aramwit et al., 2015). The kidney histopathology revealed prominent glomerular congestion, sub-capsular epithelium exfoliation, vacuolation, tubular edema, dilation, wide tubule spacing accompanied by epithelial necrosis (Fig. 6), which, however, had a specific impact on renal function as evidenced by the reduced serum CRE and BUN. Prior literature on *in vitro* and *in vivo* studies suggests that CM certainly induce toxicity in renal tubules (Chiu et al., 2016; Zhou and Yao, 2013). Moreover, due to the higher concentration of cordycepin in CM compared to *C. sinensis*, it is likely to cause nephrotoxicity (Qin et al., 2019; Yu et al., 2006). A similar concern arose in patients consuming CM enriched with cordycepin was associated with acute kidney injury (Tangkiatkumjai et al., 2022). The spleen, a vital organ of the human immune system, filters erythrocytes, eradicating antigens via phagocytosis, and defends against infections (Handy et al., 2002). The current investigation observed a substantial enlargement in the red pulp area of spleen, as evidenced by the elevated levels of RBCs, Hb and HCT, along with morphometric analysis of spleen (Fig. 7, Table. 4). Conversely, the white pulp displayed reduced lymphocytes within the PALS region, replaced by macrophages, granulocytes, and plasma cells. Besides, sinusoids exhibited minor splenic vacuolation with focal hemosiderosis in the red pulps. Nonetheless, prior research has showcased parallel findings, reinforcing immune responses (Liu et al., 2016; Zhu et al., 2013). Testis histopathological observations revealed minor changes, aligning with a sub-chronic toxicity study in SD rats (Jhou et al., 2018). CME notably influenced testicular function by increasing seminiferous tubule size and area, along with mature sperm production, as evident in histological sections (Fig. 8, Table. 5). This finding resonates with previous studies in juvenile mice and adult pigs supplemented with CME, resulting in enhanced sperm count and motility (Lin et al., 2007, 2022). Moreover, cordycepin extracted from CM improved sperm quality in aged rats (Kopalli et al., 2019). While mutagenic analysis revealed negligible anomalies in sperm morphology, CME has been linked to increase libido and fertility in male mice by raising testosterone levels, yet no such reports have been found allied to female reproductive organ (Bach et al., 2022; Hong et al., 2011). The aphrodisiac nature of CM are corroborated by various studies across different mouse age groups, indicating elevated serum testosterone and seminal parameters post-CME treatment (Chang et al., 2008; Saksopha et al., 2019; Sohn et al., 2012). Despite these positive impacts, further research is requisite to fully comprehend the precise mechanism underlying CM's effects on testicular and seminal attributes comprehensively. Ovary, on the contrary, did not exhibit any notable changes, and its histology remained unaltered compared to those of the female mice in control group. Despite an observable increase in oocyte size following the treatment, it could be contended that methanolic CME had a beneficial impact on the ovary (Fig. 9, Table. 6). This assertion can be sustained by the discoveries of Bach et al. (2022), which indicate a substantial rise in the pregnancy rates in female rats, along with enhancements in both the quality and quantity of fertility-related factors.

The sperm morphology assay is utilized to quantify mutagenic effects on male germ cells (Anderson et al., 1997). 28-day treated mice revealed minor flaws in sperm morphology, primarily head defects exceeding tail defects (Fig. 10). Interestingly, this contradicts with a study on male Wistar rats treated with aqueous CME, which exhibited a lower percentage of head defects (Tongmai et al., 2018), possibly due to extract variations, with defect percentages not significantly different from the control. Furthermore, the genotoxicity of methanolic CME was assessed via chromosomal aberrations test on bone marrow, however, they were deemed insignificant. These findings were subsequently validated by Long et al. (2021). Overall, these findings established that methanolic CME does not manifest any possible indications of sub-acute toxicity except for kidney. Our findings thus agree quite well with prior studies that corroborate the findings.

5. Conclusions

Our study determined that sub-acute treatment with methanolic CME did not elicit any drastic changes in the body weight (except for females), behaviour, and organ weight of the mice. Few haematological parameters displayed significant changes, consistent with observed histopathology alterations. Decreased serum BUN and CRE levels correlated with kidney histopathological lesions. Liver, spleen, testis, and ovary, however, exhibited normal architecture as that of control mice. CME enhanced seminal parameters and provided insights into mutagenicity through sperm morphology and chromosomal aberration studies. These findings offer preliminary insights into the toxicity of CM and underscore the need for further *in vivo* research to assess its safety and potential clinical efficacy. This endeavour could pave the way for validating its biological activities and identifying bioactive components for novel pharmaceutical development.

Ethical approval

The Institutional Animal Use and Ethical Committee (IAEC), Cotton University, and other collaborative institutes examined and approved the ethical permission towards the animal model studies (No.

14/IAEC/CU/05/01/2021). Regarding the treatment, welfare and utilization of the animals, the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA) criteria were obeyed too.

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None.

CRediT authorship contribution statement

Diksha Dutta: Writing – Original draft, Investigation, Formal analysis, Data curation. **Namram Sushindrajit Singh:** Data analysis, Visualization, Validation, Methodology, Review & editing. **Akalesh Kumar Verma:** Conceptualization, Methodology, Supervision, Validation, Review & editing.

Declaration of competing interest

The authors declare that they have no known completing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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